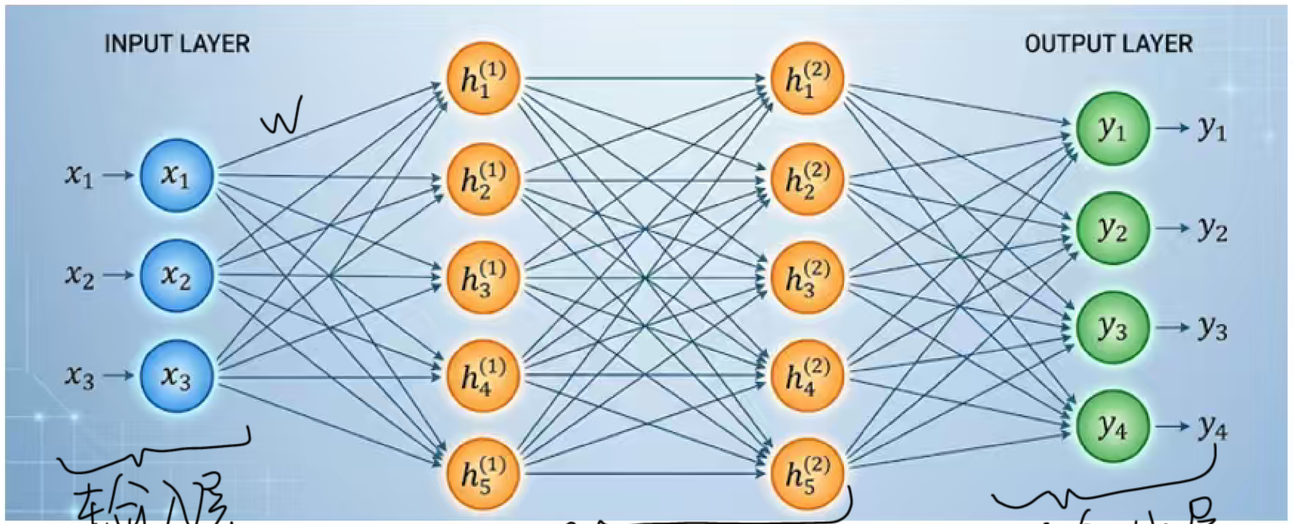
神经网络与BP算法

Back propagation

反向传播

就是个复杂的函数.

- 神经网络是一种强大的函数逼近工具，通过组合大量简单的非线性神经元，能够从数据中自动学习复杂的模式。



全连接神经网络

mmexport1776067514050.mp4
1.22 MB

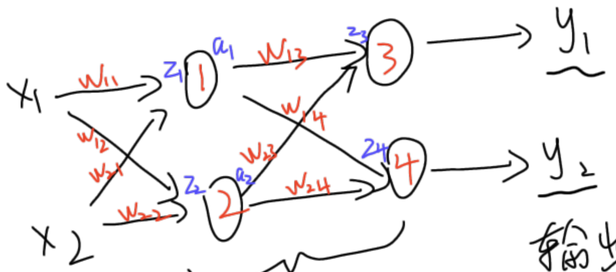
functions describe the world.

≥2 → 深度神经网络.

无论是机器学习、深度学习：从海量数据中找出规律 → 也可以看作在解一个函数.

x	y.
2	5.
3	7
4	9.

⇒ $y = 2x + 1$.



输出层

$\hat{y}_1$ L_1

$\hat{y}_2$ L_2

已知

激活函数

sigmoid函数

$\sigma(x) = \frac{1}{1+e^{-x}}$

当参数

权重 偏置

正向传播：

$z_1 = w_{11}x_1 + w_{21}x_2 + b_1$

$a_1 = \sigma(z_1) = \frac{1}{1+e^{-z_1}}$

$z_2 = w_{12}x_1 + w_{22}x_2 + b_2$

$a_2 = \sigma(z_2) = \frac{1}{1+e^{-z_2}}$

$z_3 = w_{13}a_1 + w_{23}a_2 + b_3$

$y_1 = \sigma(z_3) = \frac{1}{1+e^{-z_3}}$

$z_4 = w_{14}a_1 + w_{24}a_2 + b_4$

$y_2 = \sigma(z_4) = \frac{1}{1+e^{-z_4}}$

要计算真实值与估计值的误差：我们用 L_2 均方误差做为损失函数

$\frac{1}{n}\sum(\hat{y}_i - y_i)^2$

$L_1 = \frac{1}{2}(\hat{y}_1 - y_1)^2$ $L_2 = \frac{1}{2}(\hat{y}_2 - y_2)^2$

总损失：$L = L_1 + L_2$ → 最小化这个损失

用来示

为什么这个神经网络能表示一个非常复杂的函数呢？

$x \to \bigcirc \to y$ $y = \sigma(wx + b)$

$w = 2$ $b = 3$ $\Rightarrow \sigma(2x + 3)$

对激活函数方左右缩放平移

$x \xrightarrow{w_1} \bigcirc \xrightarrow{w_2} \bigcirc \to y$

$w_2 = 2$ $b = 2$

$y = \sigma(w_2\sigma(w_1x + b_1) + b_2)$

对激活函数上下缩放平移

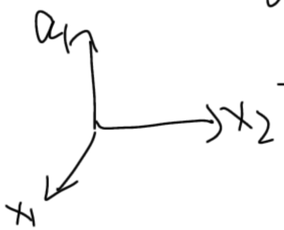
$x \xrightarrow{w_1} 1 \xrightarrow{w_3} 3 \to y$

$x \xrightarrow{w_2} 2 \xrightarrow{w_4} 3$

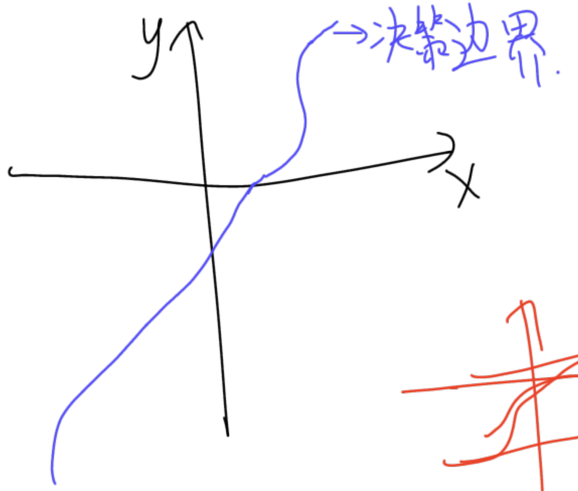
$y = \sigma(w_3\sigma(w_1x + b_1) + w_4\sigma(w_2x + b_2) + b_3)$

对两个激活函数方左上下平移与缩放后相加

$a_1 = \sigma(w_{11}x_1 + w_{21}x_2 + b_1)$

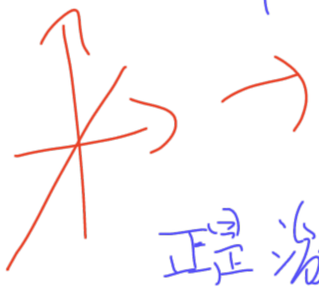


→ 三维空间缩放旋转的.



$y_1 = \sigma(w_{13}a_1 + w_{23}a_2 + b_3)$.

$= \sigma(w_{13}\underbrace{\sigma(w_{11}x_1 + w_{21}x_2 + b_1)}_{a_1} + w_{23}\underbrace{\sigma(w_{12}x_1 + w_{22}x_2 + b_2)}_{a_2} + b_3)$.



正是激活函数非线性的，才能处理非线性的任务。

即a_1在三维空间垂直缩放　　同理，a_2也一样.

然后将两个函数加起来 → 再通过函数拟合.

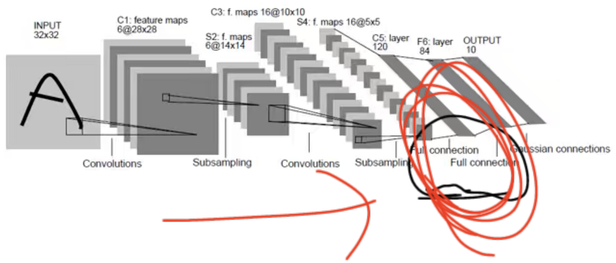
目标：通过训练获得参数，权值和偏置

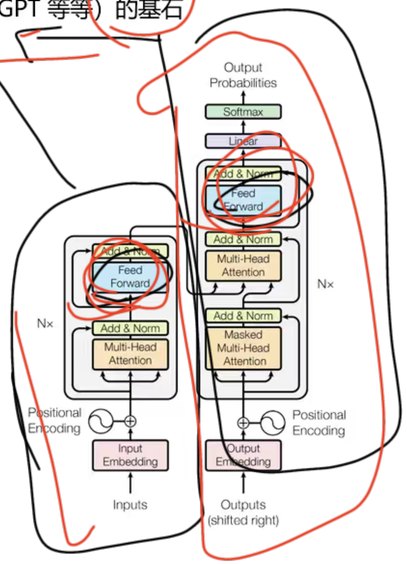
LLM → transformer

神经网络

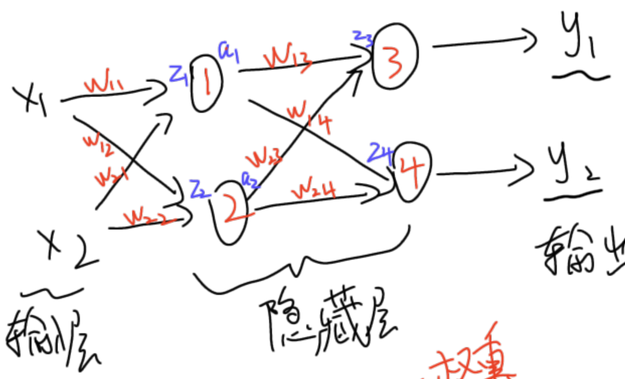
基石

- 神经网络是深度学习（CNN、RNN、Transformer、BERT、GPT 等等）的基石





那么是怎么训练的网络 → BP算法 → 训练网络



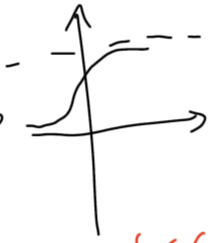
$\hat{y}_1$ L_1

$\hat{y}_2$ L_2

激活函数

sigmoid函数

$$\sigma(x)=\frac{1}{1+e^{-x}}$$



$$\frac{\partial L}{\partial w_{11}} \quad \frac{\partial L}{w_{12}} \quad \frac{\partial L}{w_{13}} \cdots\cdots$$

如果层数很深，这个计算量非常大.

$$\frac{\partial L}{\partial w_{11}}=\frac{\partial L}{\partial y_1}\frac{\partial y_1}{\partial z_3}\frac{\partial z_3}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{11}}+\frac{\partial L}{\partial y_2}\frac{\partial y_2}{\partial z_4}\frac{\partial z_4}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{11}}$$

$$\frac{\partial L}{\partial w_{13}}=\frac{\partial L}{\partial y_1}\frac{\partial y_1}{\partial z_3}\frac{\partial z_3}{\partial w_{13}}$$

$\cdots\cdots$

用链式法则

正向传播：

权重　偏置

$$z_1=w_{11}x_1+w_{21}x_2+b_1$$
$$a_1=\sigma(z_1)=\frac{1}{1+e^{-z_1}}$$
$$z_2=w_{12}x_1+w_{22}x_2+b_2$$
$$a_2=\sigma(z_2)=\frac{1}{1+e^{-z_2}}$$

$$z_3=w_{13}a_1+w_{23}a_2+b_3$$
$$y_1=\sigma(z_3)=\frac{1}{1+e^{-z_3}}$$
$$z_4=w_{14}a_1+w_{24}a_2+b_4$$
$$y_2=\sigma(z_4)=\frac{1}{1+e^{-z_4}}$$

要计算真实值与估计值的误差：我们用 L_2 均方误差作为损失函数

$$\frac{1}{n}\sum(\hat{y}_i-y_i)^2$$

$$L_1=\frac{1}{2}(\hat{y}_1-y_1)^2 \qquad L_2=\frac{1}{2}(\hat{y}_2-y_2)^2$$

总损失：$L=L_1+L_2$ → 最小化这个损失

用求导

→ 求

反向传播：

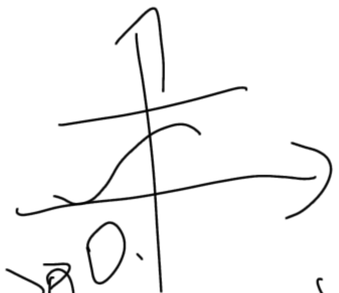
先计算对后面层的导数，从后往前，前面层计算导数就能直接用后面层计算好的导数，大大减少了计算量.

$$\underbrace{\text{正向传播} \longrightarrow \text{反向传播，求导} \longrightarrow \text{更新参数}}_{\text{一次迭代.}}$$

$$\frac{\partial L}{\partial w_{13}} = (y_1 - \hat{y}_1)\, y_1 \underbrace{(1-y_1)}_{>0} a_1$$

η 正数.

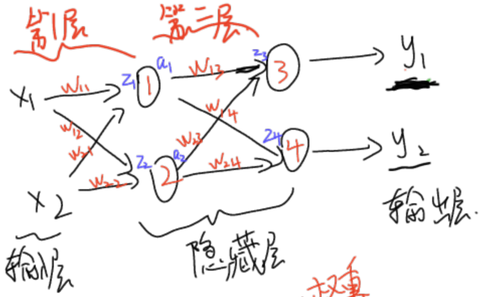
$$\begin{cases} y_1 > \hat{y}_1 \\ \text{预测大于真实} \\ y_1 < \hat{y}_1 \text{ 也是这样的} \end{cases} \longrightarrow \frac{\partial L}{\partial w_{13}} > 0 \longrightarrow \hat{w}_{13} \downarrow$$



$y_1 \downarrow$

反向传播：$\frac{\partial L}{\partial w_{13}} = \frac{\partial L}{\partial y_1}\frac{\partial y_1}{\partial z_3}\frac{\partial z_3}{\partial w_{13}} = (y_1-\hat{y}_1)\,y_1(1-y_1)\,a_1$

$\frac{\partial L}{\partial b_3} = \frac{\partial L}{\partial y_1}\frac{\partial y_1}{\partial z_3}\frac{\partial z_3}{\partial b_3} = (y_1-\hat{y}_1)\,y_1(1-y_1) = \delta^{(2)}$



$\hat{y}_1$ L_1

$\hat{y}_2$ L_2

激活函数

sigmoid函数

$\sigma(x) = \frac{1}{1+e^{-x}}$

参数更新：$\hat{w}_{13} = w_{13} - \eta\frac{\partial L}{\partial w_{13}}$ 学习率 $\quad \hat{b}_3 = b_3 - \eta\frac{\partial L}{\partial b_3}$

正向传播：

$z_1 = w_{11}x_1 + w_{21}x_2 + b_1$ （权重，偏置）

$a_1 = \sigma(z_1) = \frac{1}{1+e^{-z_1}}$

$z_2 = w_{12}x_1 + w_{22}x_2 + b_2$

$a_2 = \sigma(z_2) = \frac{1}{1+e^{-z_2}}$

$z_3 = w_{13}a_1 + w_{23}a_2 + b_3$

$y_1 = \sigma(z_3) = \frac{1}{1+e^{-z_3}}$

$z_4 = w_{14}a_1 + w_{24}a_2 + b_4$

$y_2 = \sigma(z_4) = \frac{1}{1+e^{-z_4}}$

w_{23}, w_{24} 也是这样求偏导.

$$\frac{\partial L}{\partial w_{14}} = \frac{\partial L}{\partial y_2}\frac{\partial y_2}{\partial z_4}\frac{\partial z_4}{\partial w_{14}} = (y_2-\hat{y}_2)\,y_2(1-y_2)\,a_1$$

$$\frac{\partial L}{\partial b_4} = \frac{\partial L}{\partial y_2}\frac{\partial y_2}{\partial z_4}\frac{\partial z_4}{\partial b_4} = (y_2-\hat{y}_2)\,y_2(1-y_2)$$

$$\frac{\partial L}{\partial w_{11}} = \underbrace{\frac{\partial L}{\partial y_1}\frac{\partial y_1}{\partial z_3}}_{\frac{\partial L}{\partial b_3}}\frac{\partial z_3}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{11}} + \underbrace{\frac{\partial L}{\partial y_2}\frac{\partial y_2}{\partial z_4}}_{\frac{\partial L}{\partial b_4}}\frac{\partial z_4}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{11}}$$

$$= (y_1-\hat{y}_1)\,y_1(1-y_1)\,\underbrace{w_{13}}_{\text{第二层权重}}\,a_1(1-a_1)\,x_1 + (y_2-\hat{y}_2)\,y_2(1-y_2)\,\underbrace{w_{14}}_{\text{第二层权重}}\,a_1(1-a_1)\,x_1$$

---- 其它权重也是这样计算.

公式：$\frac{\partial L}{\partial w^{(l)}} = \delta^{(l)}\,(a^{(l-1)})^T$ 上一层的输出

$\frac{\partial L}{\partial b^{(l)}} = \delta^{(l)}$

隐藏层：$\delta^{(l)} = (W^{(l+1)})^T\,\delta^{(l+1)} \odot \sigma'(z^{(l)})$

列乘

输出层：$\delta^{(L)} = \frac{\partial L}{\partial y^{(L)}} \odot \sigma'(z^{(L)})$

激活函数求导

$$\frac{\partial L}{\partial b^{(1)}} = \delta^{(1)}$$

$\delta^{(1)}$怎么算的？

$$\delta^{(1)} = (W^{(2)})^T \delta^{(2)} \odot \sigma'(z^{(1)})$$

$$\frac{\partial L}{\partial W^{(1)}} = \delta^{(1)} \cdot \underbrace{(a^{(0)})^T}_{\text{输入 } x_1, x_2}$$

$$\frac{\partial L}{\partial w_{11}} = \left[(y_1-\hat{y}_1)y_1(1-y_1)w_{13} + (y_2-\hat{y}_2)y_2(1-y_2)w_{14}\right]a_1(1-a_1)x_1$$

公式: $\frac{\partial L}{\partial w^{(l)}} = \delta^{(l)} \underbrace{(a^{(l-1)})^T}_{\text{上一层的输出}}$

$\frac{\partial L}{\partial b^{(l)}} = \delta^{(l)}$

隐藏层: $\delta^{(l)} = (w^{(l+1)})^T \delta^{(l+1)} \odot \sigma'(z^{(l)})$

列乘积

输出层: $\delta^{(L)} = \frac{\partial L}{\partial y^{(L)}} \odot \sigma'(z^{(L)})$

$\frac{\partial L}{\partial w^{(l)}}$ $\delta^{(l)} \rightarrow$ 误差项 $\underbrace{a^{(l-1)}}_{\text{上层输出}}$

$$\frac{\partial L}{\partial w^{(2)}} = \underbrace{\delta^{(2)}}_{2\times 1} \cdot \underbrace{(a^1)^T}_{2\times 1} = 2\times 2 \text{ 矩阵}$$

$$\begin{bmatrix} \sim \\ \sim \end{bmatrix} \qquad \begin{bmatrix} \sim \\ \sim \end{bmatrix} \qquad \begin{pmatrix} \bullet & \circ \\ \bullet & \circ \end{pmatrix}$$

$$\frac{\partial L}{\partial b^{(2)}} = \delta^{(2)}$$